PRATIMAI · AI HANDOUTS

How Machines Learn

My case file for training machines — and catching their mistakes

CLASS 7 · LEVEL 2 — SORT & PREDICT

17 PAPER MISSIONS · 5 CHAPTERS · NO SCREEN NEEDED

PRATIMAI Curriculum Team

How do I know? · What is missing? · What is the trick here?

THIS CASE FILE BELONGS TO

name · class · roll number

CASE OPENED ON ★ CASE CLOSED ON

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Every activity in this book is **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. All examples use invented people and places.

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Acknowledgments

To the teachers and their Classes 6 and 7 who piloted every mission with pencils, paper and great patience — your classroom notes shaped this book. You told us which tasks made the room go quiet with thinking, which ones made it erupt in argument, and where the instructions needed one more example. Every one of those notes made a page here better.

To the curriculum reviewers who guarded the "unplugged" promise on every page, thank you. You caught the three places where a task quietly expected a phone, and you were right: an investigator's tools are paper, pencils, patience and other people. And to the grown-ups at home who will be asked "how does the machine know THAT?" at the dinner table — this book is for those conversations, too. The best missions end with a child teaching an adult.

Thanks as well to the open-source community behind Typst and the Machiatto template that gives this book its typesetting, and to the type designers of Nunito and Baloo 2, whose friendly letterforms keep pages of hard thinking feeling light. Finally, to the investigator holding this book: the reviewers never got to meet you, but every confidence circle, empty line and blank chart in here was left empty on purpose — because the last author of this case file is you.

17 PAPER MISSIONS · 14 WORDS TO KEEP · 5 CHAPTERS · 1 BADGE EARNED

How to Use This Handbook

Welcome back, Detective! Last year you learned to **spot** smart machines. This year you go inside their workshop: you will find out how a machine learns **without being told the rules** — and you will train a few learners yourself. You will still not need a computer, a phone or an app. Your tools are the same as ever: paper, pencils, cards, your classmates, and your three questions. Every chapter hands you missions (we call them **tasks**), evidence to examine, and empty spaces to fill with what you discover. Good investigators write everything down, so this book belongs to you: write, draw, tick, cross out and try again.

1 READ

Read the mission twice. Underline anything that puzzles you — puzzles are clues.

2 DO

Do the task with your partner, group, class or family, exactly as the labels say.

3 WRITE

Write what you saw in the spaces, then shade your confidence circles honestly.

The work-mode labels

ALONE	your own case — you investigate by yourself
PAIR	two detectives — share one book, take turns reading codes aloud
GROUP	three to four detectives — decide roles before you start
CLASS	the whole class plays together, led by your teacher
HOME	a mission for home — show a grown-up what you discovered

HOW TO READ A MISSION

T7-01 Mission title — GROUP · 15 MIN

■ **QUICK WIN** sits at the right edge of the easier missions.

Every mission has a code like **T7-01** — **T7** means this Class 7 handout, and **01** is the mission number, so your teacher can say “open Task T7-11”. The **GROUP** label tells you who to work with, **15 MIN** is your time limit, and **QUICK WIN** marks an easy first success. Chapters 1 and 2 both open with a Quick Win to warm up your investigator muscles — nobody fails in this book, because every attempt teaches a clue.

THE QUESTION HABIT — YOUR SUPERPOWER

Three questions make you a sharper investigator than any machine: **“How do I know?”** · **“What is missing?”** · **“Who made this?”** This year they get a fourth companion you will use on every chart and every claim: **“What is the trick here?”** Use all four in this book, on the internet, in the news — and whenever a machine gives you an answer that looks too perfect.

A Note for Grown-Ups

This handout builds **AI understanding** — how machines learn from data and how humans should judge them — through 17 paper-and-pencil missions. It is fully **unplugged**: no task requires a device, an account, a photo, a voice recording or any personal information. This year the big ideas are training and testing, the three learning jobs (classification, regression, clustering), how machines see, read and recommend, charts and how they mislead, bias, fake content, and responsibility. The best help you can give is to ask the Question-Habit questions, praise careful **reasoning** more than quick right answers, and treat the confidence circles as honest thinking, not scores. A separate teacher pack carries answer notes and rubrics; this book deliberately never prints solutions beside a task — only short hints at the very back.

Four ways to help — without giving answers

1 Ask, don't tell

When your investigator is stuck, ask: **"What have you tried? What data does the machine need — and where would it get that?"** A good question unblocks; a quick answer un-trains.

2 Praise the reasoning

Swap "correct!" for **"I like how you checked that"** or **"you changed your mind when the evidence changed — that is real investigation."**

3 Let it be wrong

A confident wrong answer is useful material. Ask **"how sure are you — and what would change your mind?"** Updating a belief is the skill this book teaches.

4 Connect to real life

Point at the helpers around you — the map app, the shop scanner, the video suggestions — and ask which **three learning jobs** each one is doing.

IF YOUR CHILD ASKS…

"How does the machine learn without being told?" — Say: it is like practising questions with answers, then facing a hidden exam. Patterns it found — not rules we wrote. **"Is it always fair?"** — Say: not always. If its examples were one-sided, its answers will lean too. That is why this book trains **what is missing?** thinking. **"Who is to blame when it goes wrong?"** — Say: responsibility follows control, and machines control nothing — so the answer is always **people**: the makers, the choosers of data, the deployers.

SIGNS OF INVESTIGATOR GROWTH WORTH PRAISING

Watch for these quiet changes — they matter more than any correct answer: your child **asks how sure** they are before answering · they **check a second source** without being told · they **change their mind** when the evidence changes — and say why · they name a chart's **trick**, not just its picture · they explain a machine's **benefit AND limit** in one sentence. When you spot one, name it: "That is exactly what investigators do."

My AI Journey Map

THE FIVE STRANDS YOU WILL MEET AT EVERY STOP

U Understand AI
What are the three jobs of a learning machine?

D Data
What data does a machine need — and how do I read it?

L Learn
How does training work — and when does it fail?

W World
Where is AI working in India right now?

R Responsibility
Who is harmed when AI goes wrong — and who answers?

THINKING-TOOLS THREAD

runs through every stop: **patterns** · **if-then rules** · **elimination** · **how sure am I?**

THE FIVE STOPS ON YOUR CASE

1. **The Learning Machine**
   classification · prediction · clustering
2. **Machines That See, Read and Recommend**
   pixels · word counts · people like you
3. **AI at Work in India**
   five sectors · reading charts
4. **When AI Gets It Wrong**
   bias · fakes · responsibility
5. **Class Survey Machine**
   our own prediction machine

BY THE END OF THIS BOOK, I CAN…

- [ ] tell **classification**, **regression** (predicting a number) and **clustering** apart, with examples from my classroom
- [ ] explain training vs testing as “practice questions vs the real exam”
- [ ] explain why a few one-sided examples fool a learner — and why many varied examples help
- [ ] collect structured data with a survey, and draw and read bar, line and pie charts
- [ ] spot a misleading chart — and name the trick it uses
- [ ] explain at idea level how a machine **sees**, **reads** and **recommends**
- [ ] describe one AI helper in healthcare, education, transport, agriculture or communication — with one benefit and one limit
- [ ] explain how one-sided examples cause unfair results — and propose a fix
- [ ] apply three checks — **source**, **evidence**, **emotion** — to catch fake or edited content
- [ ] compare solving a task with fixed if-then rules vs learning from examples

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The boxes in this book

You will meet these boxes on every case. Each one wants something different from you — collect them all.

MISSION

A paper mission with a code like **T6-01**. The label tells you who you work with and how many minutes you get. Do it — then write. Writing **is** the mission.

WORD POWER

A word worth keeping in your detective kit. Say it three times today — words are the tools of thinking.

SELF-CHECK

Skills to grade yourself on. Shade a confidence circle honestly — honest circles make you learn faster.

THINK & INK

A question with lines. There is no single right answer here — your **reasons** are the answer.

Also on patrol: **MYTH BUSTER** busts a wrong idea many people believe · **DETECTIVE'S NOTE** hands you a professional's secret · **HOME LINK** (dashed) is a mission to take home and teach a grown-up.

1. The Learning Machine

GUIDING QUESTION

How can a machine learn without being told the rules?

Welcome to the workshop. Last year you learned to **spot** learning machines; this year you operate one. In five missions you will run all three learning jobs with your own hands — sorting objects into named groups (classification), drawing a best-fit line to predict a fare (regression), inventing groups nobody gave you (clustering) — and then sit the same hidden exam a machine sits, to learn how training and testing really work.

MISSIONS T7-01 – T7-05
OUTCOMES 7.U1 · 7.L1 · 7.L2 · 7.T1 **STRANDS** U L

Links: Maths — graphs & averages · English — precise definitions

In this chapter

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WORDS YOU'LL MEET HERE

classification **regression** **clustering** **training set** **test set**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

A video app always seems to know what you want to watch next. Write your best guess: WHERE did that knowledge come from? You will check this guess at the end of the book.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · ruler · 12 small objects to sort · a fair judge

1.1 Three jobs that learning machines do

Last year you met two kinds of machines: rule-followers, which repeat steps somebody wrote, and learners, which improve from examples. This year we look closer at the learners — because "learning" turns out to mean three different jobs. Some machines **put things into named groups**: spam or not-spam, cat or dog, ripe or unripe. Others **predict a number**: tomorrow's temperature, the price of a trip, the marks a form might score. And some **find groups nobody named** — they look at a pile of examples and invent their own sorting. Almost every learning machine you will ever meet is doing one of these three jobs. Today, you try all three yourself.

T7-01 Sorting Machine

QUICK WIN · GROUP · 15 MIN

Your teacher (or a detective in your group) gathers **12 small objects** — for example: a leaf, a coin, a pen cap, a chalk stub, a button, a pebble, a rubber band, a safety pin, a sticker, a clip, an eraser, a seed. Spread them out. Now be the machine: invent **labels** and put every object into a group. You may not use words like "junk" — group names must tell what the members share. Then fill the table:

Group name we invented	Members of this group	One more object we could add

Compare with another team: Did they invent the same groups? Where did you disagree?

Our cleverest group name — and why it works:

WORD POWER · 1

Definition: **classification**

Putting things into groups that already have names, like fruit or vehicles.

Say it three times today — it sticks!

DETECTIVE'S NOTE

In T7-01 **you** chose the labels first and sorted second — that is classification. A machine does it backwards: it gets thousands of examples **with** labels attached, finds the pattern, and only then sorts new things. Same job, opposite order. Keep that in mind — it will matter in Task T7-04.

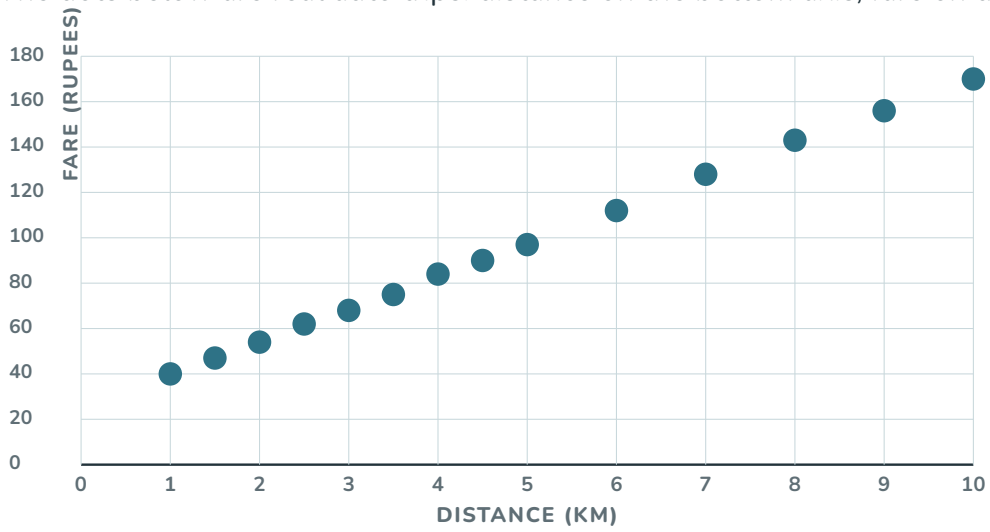
1.2 Guessing the number: the second job

Now the second job. An auto's fare grows with distance, but not perfectly — traffic, waiting and routes bend the pattern. A learning machine that predicts **numbers** from such messy data is doing the job scientists call **regression**. Nobody writes "fare = 25 + 14 per km" into the machine. Instead, it is shown hundreds of real trips, and it finds the straight line that fits them best. Your pencil can do the same thing — and yours will even show your working.

T7-02 Price Predictor

PAIR · 15 MIN

The dots below are real auto trips: distance on the bottom axis, fare on the side axis.



Step 1. With a ruler, draw the one straight line that passes as close as possible to **all** the dots. **Step 2.** Use your line to predict: the fare for a **6.5 km** trip is about ₹ _____ **Step 3.** Now a braver guess: the fare for a **12 km** trip is about ₹ _____

Why is the 12 km guess riskier than the 6.5 km guess?

Another pair drew a slightly different line. Why is that allowed — and why does the machine need one answer anyway?

WORD POWER · 2

Definition: regression

Using a pattern to predict a number, like a price or a temperature.

Say it three times today — it sticks!

1.3 Find the groups nobody named

The third job is the strangest. In classification, the groups already have names. But sometimes nobody knows the groups — a machine gets a pile of unlabelled examples and must **invent** its own sorting. Scientists call this **clustering**. Shops use it to find "crowds of similar customers". Astronomers use it to sort stars. There is a twist you must not miss: with no labels, there is often **no single right answer** — two honest machines (or two honest teams) can cluster the same things differently, and both can be useful.

T7-03 Find the Groups

GROUP · 15 MIN

Here are nine animals on cards: **eagle, penguin, bat, butterfly, crocodile, dolphin, snake, frog, sparrow**. No labels are given — that is the whole point. Sort them into groups and **invent** a name for each group. Then look at another team's table.

Our group name	Which animals are in it?	What do they share?

A group the other team made that we never thought of:

So is one of the two groupings "wrong"? What would you say to a machine that insists its clustering is the only truth?

WORD POWER · 3

Definition: clustering

Finding groups in examples that have no labels — the machine invents the groups.

Say it three times today — it sticks!

DETECTIVE DRILL · WHICH JOB IS IT?

The learning machine…	C, N or G?
sorts email into “spam” and “not spam”	
guesses tomorrow’s temperature in your town	
groups library books into similar piles, with no names given	
decides whether a photo shows a dog or a cat	
predicts how many kilograms of rice a field will give	
finds crowds of similar customers in a shop’s data	

C = classification · N = predicts a number · G = clustering

1.4 Practice questions and the real exam

How does a machine actually **learn** the pattern? Here is the secret, and you already know it from your own life: it **practises, then it is tested**. First the machine studies a **training set** — examples with the answers attached, like practice questions with the solutions in the back. It adjusts itself until it matches them well. Then comes the moment of truth: the **test set** — new examples it has never seen, with the answers hidden. If it only memorised the practice questions, the exam will expose it. A learner that scores well on practice **and** on the hidden test has truly found the pattern.

DETECTIVE DRILL · SPOT THE MEMORISER

Two students prepared for a spelling test. One learned the **rule** for adding -ing; one memorised the exact twenty words on last year’s paper. Who passes THIS year’s test — with new words? Mark each story M (memoriser) or P (pattern-finder):

The story	M or P?	What gave it away
Scores 20/20 on last year’s paper, 6/20 on this year’s.		
Scores 17/20 on last year’s paper, 16/20 on this year’s.		
Can explain WHY “run” becomes “running” but “open” stays “opening”.		

*A machine can be a memoriser too — that is exactly why honest AI builders keep a **hidden test set**. If the machine only rehearsed, the hidden exam is where it gets caught.*

T7-04 Practice vs Exam

ALONE · 10 MIN

You are the machine. Study the **practice set**: cards labelled IN or OUT.

PRACTICE SET — WITH ANSWERS

Card	IN or OUT?
12 · 15 · 21 · 9 · 33	IN
10 · 16 · 22 · 8	OUT

The pattern I learned: "A number is IN when …"

EXAM SET — ANSWERS HIDDEN

Card	My prediction
27	
14	
36	
19	
45	

Your teacher holds the answers. Check after everyone has committed — no peeking, that would be cheating the test!

My score on the exam set: _____ out of 5 If a machine scored the same, would you say it "found the pattern" or "got lucky"? Why?

WORD POWER · 4

Definition: **training set**

The practice examples a machine learns from before it meets new ones.

Say it three times today — it sticks!

WORD POWER · 5

Definition: **test set**

The hidden examples used to check whether the learning really worked.

Say it three times today — it sticks!

DETECTIVE DRILL · TRAIN AND TEST IN MY OWN WEEK

Machines are not the only learners. Name one **training moment** and one **testing moment** from your own week — moments where you practised with answers available, then faced something brand-new:

My training moment (with answers available)	My testing moment (no answers given)

*If your practice moment and your test moment were the **same questions**, you did not test — you rehearsed. Machines make this exact mistake, and so do people.*

AT A GLANCE · IF-THEN MACHINE VS LEARNING MACHINE

Ask yourself…	If-then machine	Learning machine
Where do its rules come from?	A person wrote every if-then by hand.	It found the pattern inside training examples.
What happens with a brand-new case?	It runs the same if-thens anyway.	It predicts from the pattern — right or wrong.
Can it get better at its job?	Only if a person rewrites the rules.	Yes — more varied examples can improve it.
One machine from my own day that fits		

DETECTIVE DRILL · TRAIN OR TEST?

What is happening…	Training or testing?
A cricket coach feeds 200 labelled photos of bowling actions to a machine.	
The machine now sorts 50 photos it has never seen, and we count its correct answers.	
You solve five practice sums with the answers in the back of the book.	
Your teacher gives a surprise class test with brand-new sums.	
A spam filter keeps improving as people mark more messages.	
A scientist hides 100 examples from the machine on purpose, to check it honestly.	

Why do you think the scientist hides those examples? Discuss with your partner — there is a clue in Task T7-04.

1.5 Too few examples fool everybody

Here is the trap that catches learners — machine and human alike. Show a learner only **three** examples, and it will grab the first pattern it sees, even a wrong one. Show it **twelve varied** examples, and the false shortcuts fall apart. The examples are the machine's whole world: a learner cannot ask "are you sure?" — it can only believe its data. That is why the people who build AI spend most of their time not on the machine, but on **choosing the examples**.

T7-05 Too Few Examples

PAIR · 10 MIN

Two leaf-spotting machines were trained differently. Read their training cards, then judge them.

MACHINE A — TRAINED ON 3 CARDS

Training card	Label
smooth edges · 12 cm	MANGO
smooth edges · 11 cm	MANGO
smooth edges · 13 cm	MANGO

Rule A learned: "smooth edges means mango."

MACHINE B — TRAINED ON 12 CARDS

Training card	Label
6 mango leaves: smooth · 10–15 cm	MANGO
3 jasmine leaves: smooth · 4 cm	OTHER
3 hibiscus leaves: toothed · 9 cm	OTHER

Rule B learned: "mango means smooth **and long**."

A new leaf arrives: smooth edges, only 4 cm long. Machine A says: MANGO. Machine B says: OTHER.

Who is right — and what exactly fooled Machine A?

Machine A could be fixed without any new machine. How?

MYTH BUSTER

MYTH *"AI is told every rule."*

TRUTH That is true for **rule-followers** — a lift really is told every rule. But a **learning** machine is only given examples; the pattern is found by the machine itself, and sometimes nobody can write the final rule down in full. That is exactly why people who build AI worry so much about **which examples** go into the training set — and why a learner must always face a test set before we trust it.

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can point to a classification job, a number-prediction job and a clustering job in real life ○ ○ ○
- [ ] I can explain "training vs testing" to a Class 6 student in two sentences ○ ○ ○
- [ ] I can explain why 3 examples can fool a learner and 12 varied examples are safer ○ ○ ○
- [ ] I can draw a best-fit line on dots, predict with it, and say how sure I am ○ ○ ○

THINK & INK

A machine I use guesses things about me — a suggestion, a price, a filter. Now I know its job is … (classification / regression / clustering), because …

MY THREE MACHINES — NAME THEM, JOB THEM, LIMIT THEM

A machine from my day	Its learning job (C / N / G)	One thing it got wrong or missed

DETECTIVE DRILL · PATTERN TO RULE

A fruit-sorting machine was trained on four labelled cards. You be the rule-writer: study the table, then write the if-then rule **you** would put inside an ordinary rule-following machine.

Training card	Label
heavy · smooth · yellow skin	MANGO
very heavy · rough · green skin	WATERMELON
light · smooth · red skin	APPLE
light · rough · brown shell	COCONUT

My if-then rule:

New card arrives: heavy · smooth · yellow skin. My rule says: _____ **New card arrives: heavy · rough · yellow-green skin.** My rule says: _____ — **and a learning machine might disagree! Why?**

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed machines are told every rule by their makers. Now I know the difference between a rule-follower and a learner is …

CHAPTER 1 CLUES — POCKET THESE

- Learning machines do three jobs: **classify**, **predict a number**, **find groups**.
- **Training** is practice with answers; **testing** is the hidden exam that proves the learning.
- A few one-sided examples fool any learner; **many varied examples** protect it.
- A rule-writer writes if-thens; a learner finds the rule in the data — and nobody wrote it down.

The clue I would tell my family tonight:

2. Machines That See, Read and Recommend

GUIDING QUESTION

What does a machine actually see when it looks at a picture, a message or you?

Three machines walk into your day. One **sees**: it turns pixels into features and predicts what they show. One **reads**: it counts words and frequencies — fast, powerful, and not the same as understanding. One **recommends**: it finds people whose pattern of likes matches yours and borrows their taste. In four missions you will be all three machines — with your own eyes, your own tallies, your own table of likes — and discover exactly where each one is brilliant, and where it falls flat.

MISSIONS T7-06 – T7-09
OUTCOMES 7.W1 · 7.T1 **STRANDS** W

Links: Maths — tally & frequency · Languages — idioms and meaning

In this chapter

2.1 How a machine sees 19
2.2 How a machine reads 20
2.3 How a machine recommends 22
2.4 Lost in translation 23

WORDS YOU'LL MEET HERE

feature **frequency** **recommendation** **context** **idiom**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Look up from this page for five seconds, then look back. Write three clues (features!) that told you this page is a page and not a photo of a page.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · a page of any book · a magnifying glass if you have one

2.1 How a machine sees

In Class 6 you proved that a picture is really a grid of numbers. But seeing is more than storing numbers — a useful machine must **decide** from them, and for that it computes **features**: useful clues such as “this edge is here”, “this patch is dark”, “two lines meet at a corner”. A camera does not see a cat. It sees thousands of numbers, then features, then — if enough cat-like features line up — it predicts **cat**. The picture never enters any machine the way it enters your eyes. Today you will be the camera, and you will feel exactly what is missing.

T7-06 Be the Camera

■ QUICK WIN PAIR · 15 MIN

Detective A looks at the **number grid** below — it is a photo, exactly as a machine stores it. Detective B looks only at the **empty grid**. A must describe the photo using **numbers only** — no shapes, no words like “heart” or “line”. For example: “Row 1: 0 0 0 0 0 1.” B shades the cells that A’s numbers say are 1. Then place the grids side by side.

PHOTO AS NUMBERS (A)

0	1	0	0	1	0
1	1	1	1	1	1
1	1	1	1	1	1
0	1	1	1	1	0
0	0	1	1	0	0
0	0	0	0	0	0

B’S DRAWING — SHADE THE 1s

Compare after the first try:

The numbers that helped B most were…

Second round — describe using only two features instead of every cell:

While playing A, did you ever know what the picture “was”? What does that tell you about a machine that only ever sees numbers?

WORD POWER · 6

Definition: feature

A useful clue, stored as a number, that helps a machine decide.

Say it three times today — it sticks!

T7-06 · ROUND 2 — SWITCH ROLES

Now B invents a pattern, shades it in the first grid, and describes it with numbers only. A draws what B says. Fewer numbers should be enough this time — features, not every cell.

B INVENTS ONE

A DRAWS WHAT B SAYS

B's features this time — as few numbers as possible:

Did the picture survive two round trips? What got lost on the way?

■ MYTH BUSTER

MYTH *"The computer understands pictures like we do."*

TRUTH It has never seen a picture the way you have. It matches **number patterns** — and it works astonishingly well until the world behaves strangely: a cat photographed from below, a handwritten 4 with an open top, a face in shadow. When the features stop matching, the machine fails **confidently**. You know the feeling from T7-02: a prediction can be smooth and still be wrong.

2.2 How a machine reads

Reading works the same trick — but with words. A machine cannot **understand** a message the way you do. So it counts. Which words appear, how often, in what company? **Frequency** — how many times a word shows up — turns a sentence into a row of numbers, and numbers are something a machine can compare. Your email's spam filter is doing roughly this: counting words like FREE, WINNER and URGENT, and letting the counts vote. Counting is not understanding — but you will be surprised how far it goes. And where it stops.

T7-07 Word Counter

GROUP · 15 MIN

Six messages arrived. Your group is the filter. For each message, tally the four watch-words, decide SPAM or NOT SPAM from the counts, and note the number that decided you.

Message that arrived	FREE	WINNER	CLICK	URGENT	My verdict	Count that decided
"WINNER! You are our WINNER today! CLICK here to claim your FREE prize!"						
"Team meeting moved to 3 pm. Please bring the report."						
"URGENT: CLICK in the next 10 minutes! FREE recharge for the first 100!"						
"Hello Aarav, here are the notes from today's maths class."						
"CLICK NOW! Your account is blocked! URGENT action needed!"						
"Can you send me the cricket scores from yesterday?"						

Our filter rule, written with numbers:

Write one clean message that would fool our rule into saying SPAM. What does that teach you about counting machines?

Definition: frequency

How many times a word appears — the number a counting machine reads.

Say it three times today — it sticks!

2.3 How a machine recommends

A video app suggests your next clip, a shop suggests your next book — how? Not by understanding you. By finding **people like you**: others whose pattern of choices matches yours, and collecting what **they** enjoyed. That is the whole trick. It is called a **recommendation**, and it is clustering and counting wearing a friendly face. Notice what it cannot do: it cannot ask **why** you liked something. It only knows that people with numbers like your numbers liked these things.

T7-08 People Like You

GROUP · 15 MIN

Four friends watched four films. New student Priya has seen three of them. Be the app: find Priya's closest matches, then recommend.

Viewer	Cricket Stars	Space Cats	Jungle Trek	The Lost Kite
Rohan	✓ loved	✗ skipped	✓ loved	✓ loved
Sara	✓ loved	✗ skipped	✓ loved	✗ skipped
Kabir	✗ skipped	✓ loved	✗ skipped	✗ skipped
Meena	✓ loved	✗ skipped	✓ loved	✓ loved
Priya (new)	✓ loved	?	✓ loved	?

The viewer(s) most like Priya: _____ · How did you compare? _____
Our app recommends: _____ — because the people most like Priya enjoyed it.

The app cannot ask Priya WHY she loved Jungle Trek. Name one thing about her taste that this trick can never see.

WORD POWER · 8

Definition: recommendation

A suggestion made by finding people whose pattern of likes matches yours.

Say it three times today — it sticks!

2.4 Lost in translation

Translation looks easy: swap each word for its counterpart. But you speak a language, and you know its words carry **context** — the situation around them, the jokes, the idioms. A counting machine matches patterns between languages, and most of the time it manages. Then it meets an idiom, and the pattern collapses. Try to catch it in the act.

T7-09 Translate the Tricky

PAIR · 10 MIN

A word-by-word translation machine produced these. For each, write what a human actually means.

The sentence	The machine translated…	What a human means
“It’s raining cats and dogs.”	“Water is falling. Cats and dogs are falling from the sky.”	
“Aankhon ka taara” (Hindi)	“star of the eyes”	
“When pigs fly!”	“at the time when pigs fly in the sky”	
“That test was a piece of cake!”	“the test was made of cake”	

Now your turn — an idiom from your own language:

The idiom (write it in English letters):

The machine’s word-by-word version:

What it really means:

TRANSLATOR'S TIP

A good translation machine does not translate **words** — it translates **meaning**. It reads the whole sentence (and often the ones before it) to find the context, then rebuilds the idea in the new language. Word-by-word is how a machine fails; context is how a human wins.

DETECTIVE DRILL · THE IDIOM RESCUE KIT

Idioms are word-tricks: the meaning lives in the **context around** the words, not inside them. A counting machine has no context — you do. Decode these the way a human would, then say what a machine would need to know:

The sentence	What it really means	What a machine would need to know
“Hold on a second!”		
“This bag weighs a tonne!”		
“Class, eyes here!”		

*The missing ingredient has a name: **context** — who is speaking, to whom, about what. Some translation machines learn context from millions of examples. None of them can ask you what you **meant**. That is still your job.*

DETECTIVE DRILL · WHICH NUMBERS DID THE MACHINE USE?

The machine is…	Camera, reader or recommender?
counting how often the word FREE appears	
comparing your likes with Rohan’s likes	
finding edges and corners in a photo	
noticing that customers who bought what you bought also bought this	
storing a photo as a grid of numbers, then features	
tallying which words appear in a message	

AT A GLANCE · WHAT THE MACHINE ACTUALLY SEES

THE CAMERA

photo → grid of numbers → **features** (edges, corners, colour patches) → prediction

THE READER

message → words → **counts** of watch-words → spam or not-spam

THE RECOMMENDER

your likes → find **people like you** → collect what they enjoyed

*Three different machines — the same secret. Between the machine and the world there are only **numbers**, and the numbers only matter because somebody chose which ones to count.*

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- [ ] I can explain, with numbers, how a machine “sees” a picture — without saying it understands one ○○○
- [ ] I can build a counting rule that separates spam from not-spam, and name one way to fool it ○○○
- [ ] I can find the “people like me” in a likes table and predict what they would recommend ○○○
- [ ] I can explain why word-by-word translation fails on idioms, with one example ○○○

THINK & INK

One recommendation I received this week (a video, a song, a product) was… I now know it appeared because …

MY WEEK OF MACHINES — SPOT ALL THREE JOBS

When	The machine I met	Its job: SEE / READ / RECOMMEND	One limit I noticed
this morning			
this evening			
tonight			

*Bonus detective move: for any READ machine above, write the one **watch-word** it probably counts — then imagine a message that fools it.*

HOME LINK *take this page home — no screen needed*

AT HOME Spam Hunt

HOME · 15 MIN

With a grown-up, look at the last five messages a family group-chat received (do not open links, do not tap anything). Tally: how many carry a watch-word like FREE, WINNER, URGENT or CLICK? How many try to make you **feel** something before you **think**?

Watch-words found:

Messages that pushed an emotion first:

One message our family should stop forwarding, and why:

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter I believed machines that talk, see and suggest must understand us. Now I know they are really doing three things with numbers:

DESIGN DRILL · INVENT A NEW FILTER

Every counting filter has a job. Invent one for your classroom — a **homework-excuse detector**, a **kindness filter** for the class wall, or your own idea. Then break it.

My filter's name and job	My watch-words	My counting rule	One honest way it fails

*Every filter fails somewhere — knowing **where** is what makes you the responsible designer.*

CHAPTER 2 CLUES — POCKET THESE

- A camera sees **numbers and features** — it never sees a picture the way you do.
- A reader **counts words** — counting is powerful, but it is not understanding.
- A recommender finds **people like you** and borrows their taste — it cannot ask you why.
- Translation fails on **idioms and context** — the meaning lives around the words, not inside them.

The clue I would tell my family tonight:

Chapter 2 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. A machine reads "I could KILL for a glass of water!" and flags the message as violent. What is missing from its reading?

2. Your video app recommends cricket clips all week. Whose pattern of likes is it actually borrowing?

3. Name one job where counting words is enough — and one job where counting is not enough.

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT SEEING, READING AND RECOMMENDING

3. AI at Work in India

GUIDING QUESTION

Where is AI already working — and how do I read its report card?

Learning machines are already your neighbours' colleagues. In this chapter you will tour five sectors — healthcare, education, transport, agriculture, communication — and give each helper a report card with a benefit AND a limit. Then you will learn their favourite language: charts. You will collect structured data, draw bar, line and pie charts, and then face the chart detective's hardest truth: some charts are drawn to mislead you.

MISSIONS T7-10 – T7-12
OUTCOMES 7.W2 · 7.D1 **STRANDS** W D

Links: Maths — data handling & percentages · Social Science — community work

In this chapter

3.1 Five sectors, five helpers 29
3.2 Every helper has a benefit and a limit 30
3.3 Reading the report card: bar, line, pie 30
3.4 When a chart lies 33

WORDS YOU'LL MEET HERE

structured data **trend** **sample** **benefit** **limit**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Name one machine you have seen helping an adult do their job — at a shop, a clinic, a bus stand, a farm or a bank. What did it seem to be doing?

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · ruler · colours for charts · your sharpest suspicion

3.1 Five sectors, five helpers

Learning machines are already at work across India — not as magic, but as helpers with a job description. In **healthcare**, a machine scans lakhs of eye photos and flags the ones a doctor should check first. In **education**, an app watches which sums you get wrong and serves practice that fits you. In **transport**, map apps learn traffic patterns and reroute your bus. In **agriculture**, a farmer photographs a diseased leaf and gets a probable diagnosis. In **communication**, translation and voice tools carry a sentence across languages. Notice the pattern in every story: **data in, prediction out, then a human decides what to do**. The machine is the assistant — the judgement stays with people.

T7-10 One Problem, Many Data

GROUP · 15 MIN

A team wants to build a **crop-disease helper**: a farmer photographs a leaf, the machine predicts the disease. Every good machine begins with this question — **what data would it need to learn?** Fill the table for at least four kinds of data.

Data the machine would learn from	Where could it come from?	Who would collect it?	One care or limit

Suppose 90 of the 100 leaf photos come from one district. What problem is already growing inside this data?

The photos show farmers' fields. Should the team ask permission first? Why?

WORD POWER · 9

Definition: **structured data**

Facts arranged in rows and columns, so patterns can be found.

Say it three times today — it sticks!

A HEALTHY DATA DIET

You already know the Class 6 rule: **garbage in, garbage out.** Structured data makes patterns findable — but only **honest, varied** data makes them true. Before trusting any table, a data detective asks: Who collected this? Who is missing from it? And would the collector profit from a particular answer?

3.2 Every helper has a benefit and a limit

No helper is magic, and honest reporting means saying both halves out loud: what it does well, and where it can fail. An eye-screening machine can read photos all night without tiring — but it was trained on certain cameras and certain eyes, and it cannot explain its decision. A translation app carries your sentence across languages — but you met its downfall in T7-09. Being AI-literate is not about being amazed or alarmed; it is about holding both facts at once.

T7-12 Benefit–Limit Cards

PAIR · 10 MIN

For each sector, write **one** real benefit and **one** honest limit. No repeats across rows.

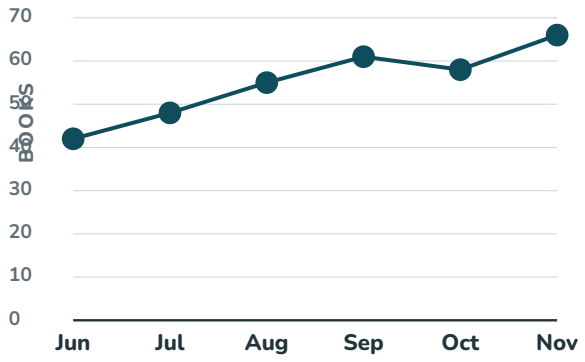
Sector	One benefit it brings	One honest limit
Healthcare		
Education		
Transport		
Agriculture		
Communication		

Which limit worries you most — and what would you tell the people who built it?

3.3 Reading the report card: bar, line, pie

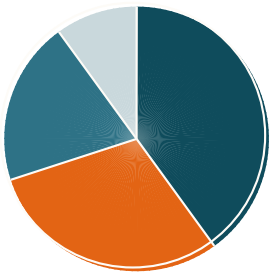
How do we know whether an AI helper — or any claim — is doing well? Its **data** is the report card, and the report card speaks through charts. You own the bar chart already: it **compares groups**. The **line graph** adds time: its slope tells the **trend**, the general direction — rising, falling, holding steady. The **pie chart** shows one whole split into parts. Each chart answers a different question, and a chart detective knows which is which: **“How many in each group?”** — bars. **“How is it changing?”** — line. **“What are the parts of one whole?”** — pie.

LINE GRAPH · BOOKS BORROWED, JUNE–NOVEMBER



The **trend** is clearly _____ — except a dip in _____ .

PIE CHART · HOW CLASS 7B TRAVELS TO SCHOOL



- Walk — **40%**
- Bus — **30%**
- Cycle — **20%**
- Car — **10%**

The whole circle is one class. Which answer can a pie chart **not** give you — the biggest group, or the exact number of children?

WORD POWER · 10

Definition: trend

The general direction a line graph shows: rising, falling or steady.

Say it three times today — it sticks!

WORD POWER · 11

Definition: sample

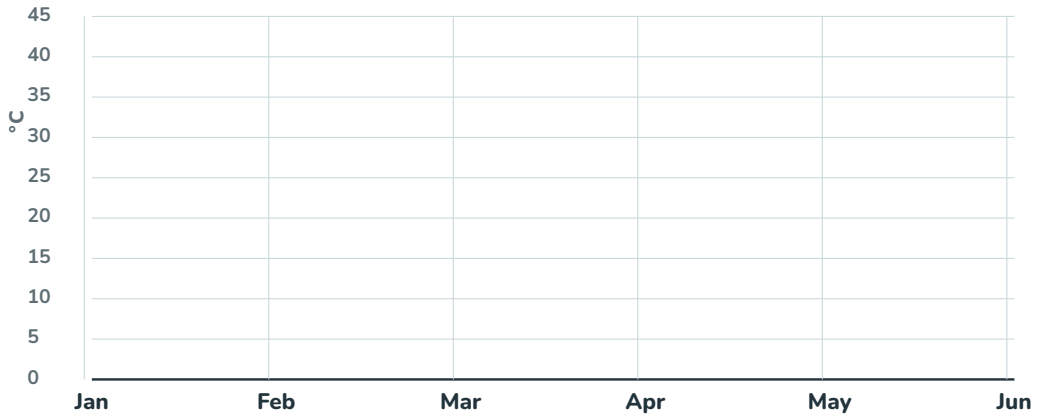
The group you actually collected data from — always smaller than everyone.

Say it three times today — it sticks!

DETECTIVE DRILL · DRAW THE TREND YOURSELF

A weather station logged the maximum temperature (°C) on the first of six months:

Month	Jan	Feb	Mar	Apr	May	Jun
"Temperature (°C)"	"18"	"22"	"28"	"36"	"40"	"38"



Hottest first-of-month: _____ · Describe the trend from Jan to May, then what changed in Jun:

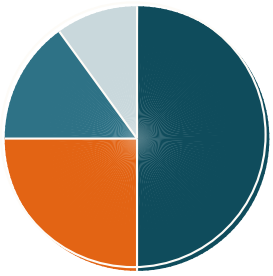
_____

DETECTIVE DRILL · WHICH CHART WOULD YOU CHOOSE?

The question…	Bar, line or pie?
Which of five house teams collected the most bottle caps this month?	
How did our school's electricity bill change from June to December?	
Of the 40 books our class read, what share were folk tales?	
How does a sapling's height change day by day?	
Which canteen item — samosa, juice, banana or sandwich — is most popular?	
How does one whole day of a doctor split across ward rounds, files, surgery and rest?	

Bars compare groups · lines follow change over time · pies split one whole into parts.

DETECTIVE DRILL · READ THE PIE YOURSELF



Library week audit — 80 books borrowed.

Roughly how many of the 80 were folk tales? _____

Which two kinds together take about one quarter of the circle?

_____

Can the pie tell us whether science books went UP this term? Why not?

PRACTICE THE SURVEY — POLITELY

Next chapter your class builds a survey machine. Practise the manners now: ask **exactly** the question as written (no hints, no leading), thank every answer, and record the answer you **heard** — not the one you **hoped for**. A survey is only as honest as its collector.

3.4 When a chart lies

Charts look like evidence — but a chart is a **drawing made by a person**, and a person can make choices that mislead you: an axis that starts high instead of zero, percentages that add up to more than 100, or numbers quietly left out. The colour and the confidence stay; the truth goes missing. In the next mission, one of the three charts is honest and two are tricksters. Read every number before you believe any picture.

THE HONEST-CHART PROMISE — WHAT A FAIR DRAWER ALWAYS DOES

An honest chart…	Tick when you check it
starts its number axis at zero — or says clearly where it starts	
makes percentages add up to exactly 100	
says where the data came from (the sample) and when it was collected	
matches the question: bars to compare, a line for time, a pie for one whole	

*Cut this promise out (or copy it) and keep it in your notebook. Any chart that breaks a promise is not lying to your eyes — it is lying to your **reason**. Detectives read numbers first and colours second.*

T7-11 Chart Detective

ALONE · 15 MIN

Three posters went up on the school noticeboard. One is honest. Two are playing tricks.

CHART A · FAVOURITE FRUIT, 7C

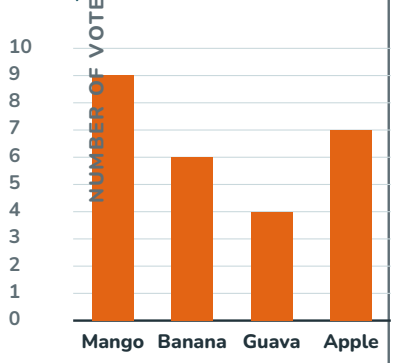


CHART B · ATTENDANCE %, 7A–7D

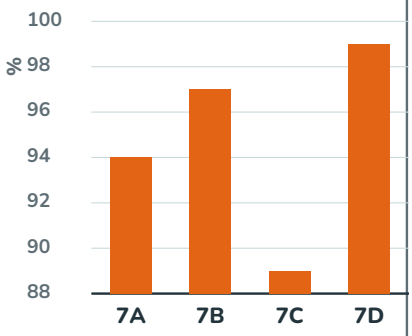
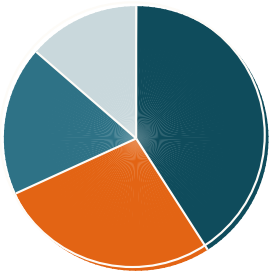


CHART C · WHERE SCHOOL MONEY GOES



- Rooms — **45%**
- Library — **30%**
- Sports — **20%**
- Computers — **15%**

The honest chart is: _____ — because…

Chart B's trick:

Chart C's trick (check the numbers like a detective):

Redraw plan — one change that would make each trickster honest:

MYTH BUSTER

MYTH *"If the chart is colourful, it is true."*

TRUTH Colour is a costume, not evidence. A chart can be beautiful and still be built on a chopped axis, a one-sided sample or numbers that never add up — you caught two doing exactly that in T7-11. The detective's move is to read the **numbers first** and the picture second: where does the axis start, what is the sample, and what is missing?

HOME LINK *take this page home — no screen needed*

AT HOME **Chart Hunt** HOME · 15 MIN

Find one chart this week — in a newspaper, a textbook, a shop poster or a government notice. Bring it (or copy it) and interrogate it: **What does it claim?**

Where does the axis start — and what is the sample?

One thing that is missing:

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- ☐ I can name one AI helper in each of the five sectors, with a benefit and a limit for two of them ○○○
- ☐ I can choose the right chart — bar, line or pie — for a question, and say why ○○○
- ☐ I can spot a truncated axis or a broken pie, and explain the trick it plays ○○○
- ☐ I can name the sample behind a chart and say who is missing from it ○○○

THINK & INK

A chart or "report" I believed without checking was… Now, before I believe a chart, I will first look at …

CHAPTER 3 CLUES — POCKET THESE

- Every AI helper is **data in, prediction out — then a human decides**. Its benefit and limit come as a pair.
- **Bars compare groups · lines follow time · pies split one whole.** Pick the chart that matches the question.
- Read a chart's **numbers first**, colours second — check where the axis starts and what adds up.
- Every chart has a **sample** — and someone is always missing from it.

The clue I would tell my family tonight:

Chapter 3 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Name one AI helper from this chapter, its benefit — and one limit nobody should forget.

2. A pie chart shows three school clubs at 60%, 30% and 40%. What is wrong — and what might the honest numbers be?

3. Your class surveys only the cricket team about favourite sports. What is the problem with that **sample**?

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT CHARTS AND HELPERS

4. When AI Gets It Wrong

GUIDING QUESTION

Who taught the machine — and who answers for its mistakes?

Every machine in this book can fail — and this chapter is about **how** and **who answers**. You will train a deliberately one-sided suggestion machine and watch unfairness walk in with the data. You will run the three checks on content that was faked or edited, and hunt what is missing. Then you will take on the hardest question of all: when a machine causes harm, who is responsible? Spoiler: responsibility follows control — and machines control nothing.

MISSIONS T7-13 – T7-15
OUTCOMES 7.R1 · 7.R2 **STRANDS** R

Links: Social Science — fairness & rules · English — checking evidence

In this chapter

4.1 One-sided examples, unfair machines 38
4.2 Fake and edited content: three checks 40
4.3 Who is responsible? 41
4.4 The fairness toolkit 43

WORDS YOU'LL MEET HERE

bias **misinformation** **verify** **fairness** **accountability**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

A school team photo shows only the winning team. Would that photo be fair evidence of "our school loves cricket"? Write what is missing.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · a calm head · courage to disagree politely

4.1 One-sided examples, unfair machines

In Class 6 you taught an alien the idea of **mango** using only green mangoes — and the alien concluded that all mangoes are green. Machine A in T7-05 fell for three leaves. Here is the grown-up version of the same trap: when the **examples behind a machine** are one-sided, the machine's predictions treat everyone outside that side unfairly. Nobody **programmed** the unfairness. It walked in quietly, inside the data. This is called **bias**, and a detective who understands it can spot it — and propose a fix.

T7-13 The Unfair Machine

GROUP · 20 MIN

A company built a **Uniform Suggester**: upload a school photo, and the app confirms your school's uniform. It was trained on 5,000 photos collected from three nearby schools — every one of them wears a **white shirt and grey trousers or skirt**. Then students of Riverdale School, whose uniform is a **blue pinafore**, tried the app.

Question	Your group's answer
What did the app predict for Riverdale students — and why?	
Who is harmed, and how? (list every group you can find)	
Who is NOT harmed — and might never notice the problem?	
One fix the company could make	
Where the better training data could come from	

Connect the cases: what does the Uniform Suggester share with the alien's green mangoes and Machine A's three leaves?

Write one honest sentence the company could send to Riverdale School:

Definition: bias

When examples are one-sided, so the machine's results treat some people unfairly.

Say it three times today — it sticks!

ONE MORE CASE · THE TALENT FINDER

A sports academy built a **Talent Finder**: feed it a video of a child playing, and it predicts who should be invited for trials. Its training data: 3,000 video clips of children playing tennis-ball street cricket — collected from one big city, all of them boys. Then two children sent videos: **Meera**, who plays handball, and **Joseph**, who plays hockey on a village field.

Child	What the machine predicted	Why did it fail?	One fix
Meera	"no talent found"		
Joseph	"no talent found"		

Who is missing from this training set? List every group you can find:

DETECTIVE DRILL · SPOT THE MISSING

The system was trained on…	Who is missing — and what could go wrong for them?
A voice assistant trained only on adult voices	
A bus-route predictor trained only on Sunday traffic	
A spelling app trained only on English words	
A face-unlock system tested mostly on a few local families	
A homework helper trained only on one board's textbooks	

DETECTIVE'S NOTE

Notice something important: the app was **not told** to prefer white shirts. Bias is not usually a bad decision somebody made — it is a **decision nobody noticed they made**: which schools, which cameras, which streets, which names ended up in the training set. That is why fairness must be **checked**, not assumed.

4.2 Fake and edited content: three checks

Class 6 gave you two checks for suspicious content: **who made this?** and **what is missing?** This year, three sharper checks — because fakes have grown up too. Check the **source**: who first published it, and are they named? Check the **evidence**: is there a photo, a document, a number you could verify somewhere else? Check the **emotion**: is the message built to make you panic, laugh or rage **before you think**? Strong emotion is not proof of a lie — but it is the costume that most lies prefer. And remember from Chapter 2: a machine that **counts** words can be taught to spread them, too.

T7-14 Fake Check 3

PAIR · 10 MIN

Three messages landed in the class group. Run all three checks on each, then give a verdict: **forward**, **stop**, or **ask a trusted adult first**.

The message	Source check	Evidence check	Emotion check	Verdict
"FREE recharge for the first 100 students! Forward to 5 groups within 10 minutes!!!"				
"District science fair moved to Saturday, 10 am — circular signed by the Principal, posted on the school website."				
"SHOCKING! Famous actor endorses magic herb that cures all fevers in one day!!"				

Which check caught the most? _____ · **Write the rule in your own words:**

WORD POWER · 13

Definition: misinformation

False or edited content that spreads — sometimes by mistake, sometimes on purpose.

Say it three times today — it sticks!

THE THREE CHECKS, APPLIED — A WORKED EXAMPLE

"Parents: the district is cancelling next Friday's exam! I know someone in the education office. Share with every class group NOW!"

Check	What the detective found
SOURCE	No name, no office, no notice number — "I know someone" is not a source.
EVIDENCE	No circular, no website, no letterhead to verify anywhere.
EMOTION	Urgency (NOW!), fear of exams, and flattery of parents — pressure before thought.
VERDICT	STOP. Ask the class teacher before forwarding anything.

DETECTIVE DRILL · WHICH BUTTON IS IT PUSHING?

The message shouts…	Fear, greed, anger — or none?
"Your account will be CLOSED in 24 hours!"	
"You have WON a free bicycle — claim today only!"	
"They don't want you to know this truth about exams…"	
"The library will stay closed on Sunday for repairs."	

Strong emotion is not proof of a lie — but it is the costume most lies prefer. When a message pushes a button, press pause instead.

4.3 Who is responsible?

A homework-help app gives a wrong answer. A student copies it. A teacher marks it right without checking. A company sells the app "100% correct". When the mistake is discovered, watch the parcel fly: the company points at the users, the student points at the app, the app's makers point at the training data. Machines cannot hold responsibility — only **people** can. So the detective's question is not "who is guilty?" but "**who controlled what?**" — because responsibility follows control.

T7-15 Who Is Responsible?

CLASS · 15 MIN

The class becomes the review panel. Four roles — **app maker, student, school, app company** — each argue their share of responsibility for the wrong answer that got submitted. Then the class votes.

Role	What this role controls	Their share (shade the bar)	One action they must take
App maker			
Student			
School			
App company			

After the vote: could any role say "the machine did it, not me"? What did the class decide, and why?

WORD POWER · 14

Definition: **accountability**

Being answerable for what a system does — only people can hold it.

Say it three times today — it sticks!

THE MAKER writes the machine, chooses the training data, tests it before release

THE USER decides when and how to use the machine — and when NOT to

THE SCHOOL sets the rules for what may be used for homework and how

THE COMPANY sells it, advertises it, profits from it, must answer for harm

Responsibility follows control: the more a role controls, the more it must answer for. And the machine controls nothing — it only predicts.

MYTH BUSTER

MYTH *"AI is neutral and objective."*

TRUTH A machine does not take sides the way people argue — but "neutral" is a costume too. Its training data was chosen by people; those choices carry the streets, schools and faces somebody thought to include — and the ones they forgot. A neutral machine over one-sided data is just bias with better manners. The fix is not magic; it is what you proposed in T7-13: wider examples, honest checks, and people who stay answerable.

4.4 The fairness toolkit

You now own three fixes, and they work on any machine, not just the ones in this book. **Widen the examples:** make the training set look more like the whole world it must serve — new schools, new games, new voices, new streets. **Run honest checks:** test the machine on the very groups it was trained without, and publish what you find. **Keep people answerable:** a person — named, reachable, responsible — must stand behind every prediction. None of these needs a computer. All of them need someone who refuses to look away. That someone is you.

WIDEN THE EXAMPLES
Train on everyone. Before judging a machine, ask: who is inside this data — and who never got in?

RUN HONEST CHECKS
Test on the missing groups. The true test set includes the people the training set forgot.

KEEP PEOPLE ANSWERABLE
Name the human. Every AI system needs a person who must answer when it goes wrong.

MY FAIRNESS PLAN

One machine I use could be unfair to:

The group missing from its training data might be:

My fix, using the toolkit:

HOME LINK *take this page home — no screen needed*

AT HOME Three Checks at the Shop

HOME · 10 MIN

Find one **claim on a packet, poster or advertisement** at home — something that promises a result ("whiter in 3 days!", "9 out of 10 doctors…", "extra power!"). Run the three checks with a grown-up: v(3pt) [**The claim:**

] [**Source — who says so, and do they profit?**

] [**Evidence — what proof is printed?**

] [**Emotion — what is it promising or frightening?**

]

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- ☐ I can explain how one-sided training data becomes an unfair machine, with a real example ○○○
- ☐ I can run the three checks — source, evidence, emotion — on a message I receive ○○○
- ☐ I can name what each role controls when an AI system causes harm ○○○
- ☐ I can explain why "the machine did it" can never be the end of the story ○○○

THINK & INK

If a machine treated someone unfairly, I used to think the fix was... Now I think the fix begins with ...

CASE NOTES — WHAT CHANGED IN MY THINKING?

Before this chapter, if a machine was unfair I would have blamed ... Now I would first look at ...

CHAPTER 4 CLUES — POCKET THESE

- **One-sided training data** builds unfair machines — nobody programmed the unfairness, it walked in with the data.
- Three checks before forwarding: **source · evidence · emotion**. Strong feeling is a costume, not proof.
- **Responsibility follows control** — and the machine controls nothing, so it can never be the final answer.
- The fairness toolkit: **wider examples · honest checks · a named human who answers**.

The clue I would tell my family tonight:

BEFORE THE CHECKPOINT — FLIP BACK TO

the Uniform Suggester cards (for clue 1) · the faked-video card (for clue 2) · the Talent Finder case (for clue 3). Each checkpoint answer should name a page, a card or a person — vague memories score zero in real investigations too.

Chapter 4 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. A uniform suggestion machine was trained only on monsoon photos. What will it suggest in May — and why is that unfair?

2. A video says “share before it is deleted!!!” Which of the three checks does it attack first?

3. Why can a machine never be the **final** answer to “who is responsible?”

MY CASE JOURNAL — TODAY'S SHARPEST CLUE ABOUT FAIRNESS AND RESPONSIBILITY

MY FAIRNESS PLEDGE — SIGN IT LIKE A PROFESSIONAL

As an investigator of learning machines, I promise to: ask **who chose the examples** before I trust a result · run the **three checks** before I forward anything · look for **who is missing** from every sample · and remember that **responsibility follows control** — so I will name the people, never “the machine”, when something goes wrong.

INVESTIGATOR’S NAME _____ DATE _____ WITNESS (A CLASSMATE) _____

5. Class Survey Machine

GUIDING QUESTION

Can our class build — and check — its own prediction machine?

The case that clicks the whole book together. Your class becomes the machine: collect structured data with a real survey, chart it, read the pattern, make a prediction — then do what honest machines must do: check your own sample for bias, predict a number from a trend, and grade yourselves against a test set from another section. Three periods, one machine, zero magic — and a badge at the end.

MISSIONS T7-16 – T7-17
OUTCOMES 7.U1 · 7.L1 · 7.L2 · 7.D1 · 7.W1 · 7.W2 · 7.R1 · 7.R2 · 7.T1 **STRANDS** U D L W R

Links: Maths — data & trends · Social Science — surveys · English — reporting

In this chapter

5.1 The whole case, in one machine 48
5.2 One more loop: predict a number 51
5.3 Reflection passport 53

WORDS YOU'LL MEET HERE

structured data **sample** **trend** **prediction** **report card**

WARM-UP · BEFORE YOU READ ALONE · 2 MIN

Your class is about to become a machine. Which job sounds most like you: asking questions (sensor), writing answers down (memory), drawing the chart (display), or checking for mistakes (testing)? Circle one — jobs come next.

DETECTIVE'S TOOLKIT — YOU WILL NEED: pencil · the survey pages · another section of Class 7 (the test set) · fair-play manners

5.1 The whole case, in one machine

This is the mission where the whole book clicks together. Your class will build a **prediction machine** out of yourselves: collect structured data, chart it, read the pattern, make a prediction — and then do the one thing most people skip: **check your own sample for bias** before trusting the prediction. Three periods. Every job you did in Chapters 1–4 comes back once: classification (Step 1), structured data (Step 2), charts (Step 3), patterns and regression-thinking (Steps 4–5), bias (Step 6), and accountability — because at the end, **your machine** must give an honest report card (Step 7).

YOUR MACHINE'S SIX PARTS — THE FULL LOOP AT A GLANCE

1 COLLECT
ask a real question, gather answers from a real sample

2 CHART
draw the data so eyes can read it: bars, pie

3 PATTERN
say what repeats — with numbers, not vibes

4 PREDICT
extend the pattern to a case you have not seen

5 CHECK
interrogate your own sample: who is missing?

6 GRADE
face the test set and grade yourselves honestly

Keep this row of parts in sight: every task in this chapter fills one of these six boxes — and the CHECK part is the one most people skip.

CHOOSE THE QUESTION — WHAT WILL OUR MACHINE PREDICT?

A machine is only as good as its question. Draft two candidate questions below, then test each against the honesty checklist and circle your winner:

Candidate question for our class survey	Answers it could offer (A–D)	Final answer? (yes/no)

The honesty checklist: exactly 3–4 possible answers · nobody will feel judged by any answer · every classmate can answer from their own day.

ROLES **Asker** keeps the wording exact · **Tally-keeper** writes counts · **Skeptic** hunts bias · **Chartist** draws — pick one and own it.

T7-16 Class Survey Machine · Part 1 — Collect and Chart

PROJECT · 3 PERIODS MIN

1 **CHOOSE.** One question with 3–4 short options. Example: **"How do you reach school on most days?"** — A Walk · B Cycle · C Bus · D Car. Or invent your own question about reading, food, games or water.

Our question:

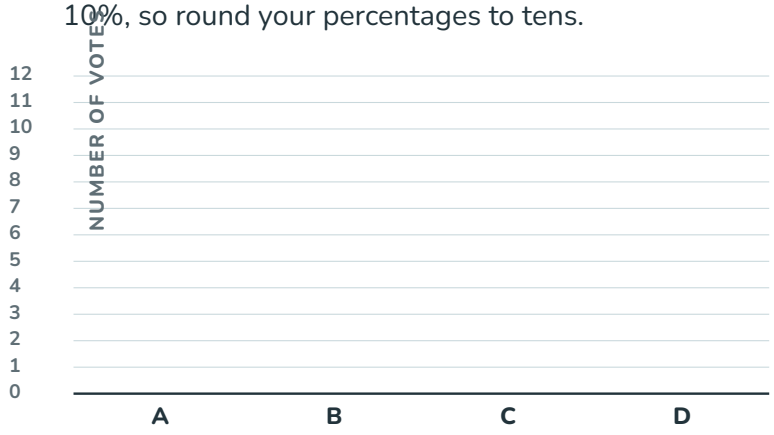
Our options: A _____ B _____ C _____ D _____

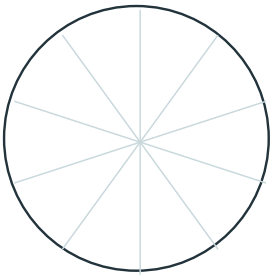
2 **COLLECT.** Ask every member of your class. Tally honestly — no votes for the option you **hope** wins.

Option	Tally marks	Count
A _____		
B _____		
C _____		
D _____		

Our sample: _____ students of _____ in the whole class. **Who is missing from our sample, and why?** _____

3 **CHART.** Draw a bar chart of your counts. Then shade a pie chart — each wedge below is 10%, so round your percentages to tens.





T7-16 Class Survey Machine · Part 2 — Predict and Check

PROJECT · CONT. MIN

4 **READ THE PATTERN.** Say it as a machine would — with numbers.

The pattern I see:

5 **PREDICT.** Another section of Class 7 will answer the same question tomorrow. Predict their top answer from your pattern.

I predict the other section's top answer will be: _____ — because the pattern says

How sure am I? ○○○

6 **CHECK THE SAMPLE.** This is where honest machines are made. Run the bias checks on your own data before you trust your prediction.

Was our sample one-sided — all same age, same class, same street?

What could that one-sidedness do to our prediction?

7 **MACHINE REPORT CARD.** Now the test set: ask the other section. Compare, then grade yourselves — honestly, like a real test.

Their actual top answer: _____ · Our prediction was _____ (right / close / off)

Our machine's honest verdict — what worked, what we would train better next time:

MYTH BUSTER

MYTH *"More data always fixes it."*

TRUTH Only **more varied** data fixes it. Collect ten thousand more forms from the same three schools and the Uniform Suggester grows **more** confident in its mistake — louder data, same blindness. What heals a learner is variety: new schools, new streets, new faces. When you meet a claim built on "huge amounts of data", ask your sharpest question: **varied for whom?**

5.2 One more loop: predict a number

Your survey machine predicted an **answer** — classification work. Real machines also predict **numbers**, and so can yours. Here is a whole week of noon temperatures your class logged on the terrace. Read the trend, then do what a regression machine does: draw the pattern forward and predict — and say how sure you are.

DETECTIVE DRILL · REGRESSION NEAR HOME

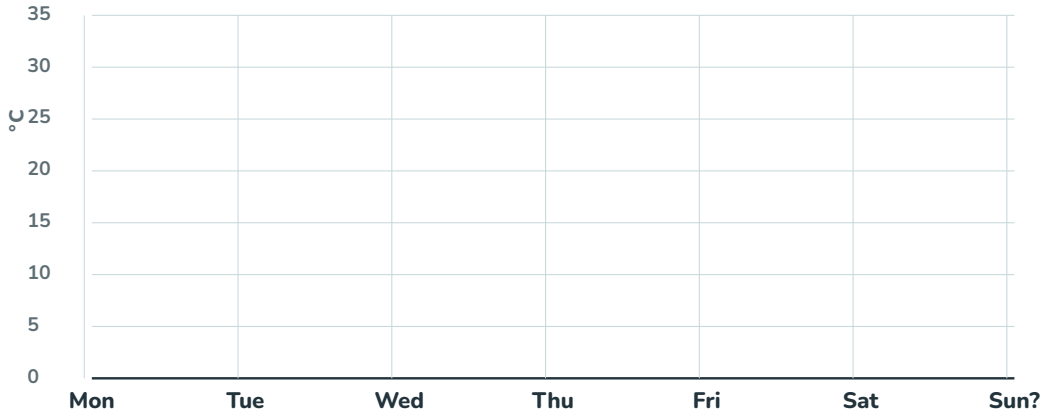
You already reason with trends every week. Extend each pattern like a regression machine would — write your number, then shade how sure you are:

The pattern so far	My prediction	How sure?
An auto charged ₹40 for 1 km, ₹55 for 2 km, ₹70 for 3 km. Fare for 4 km?	u2003	○○○
The water tank had 90 L on Monday, 80 L on Tuesday, 70 L on Wednesday. Litres on Friday?	u2003	○○○
A mango tree gave 12, then 15, then 21, then 26 mangoes each week. Next week?	u2003	○○○

Watch the third row: living things grow less neatly than meter readings. A good investigator widens the confidence circles when the data gets livelier — exactly what real regression software does.

DETECTIVE DRILL · PREDICT TOMORROW

Day	Mon	Tue	Wed	Thu	Fri	Sat
“Noon temperature (°C)”	“24”	“26”	“27”	“29”	“30”	“32”



Plot the six days, then continue the pattern to Sunday.

My prediction for Sunday: _____ °C · **How sure am I?** ○○○

What could make the real Sunday break the pattern?

If Monday suddenly showed 18 °C, what should a good machine do — keep predicting, or ask for help? Why?

THE SUNDAY CHECK — grade your own machine. When Sunday arrives, write the real noon temperature here: _____ °C. Off by 2 °C or less: your trend earns a star. Off by more: name one thing the week’s data never told you — that is the missing feature.

DETECTIVE DRILL · TREND OR NO TREND?

A trend is only worth predicting from when it holds **long enough and steady enough**. Mark each data story T (trend — safe to extend carefully) or N (no trend — a guess dressed as a pattern):

The data story	T or N?	My reason in five words
Six days of noon temperature, rising smoothly from 24 °C to 32 °C.		
Three dice throws: 6, 6, 6.		
The bus has arrived within two minutes of time every day this month.		
My marks went up after I started practising daily — five tests in a row.		

*The dice are the impostor to watch: three sixes FEEL like a pattern but carry no cause behind them. A real trend has a **why** — sunshine, practice, a timetable. No why, no prediction.*

5.3 Reflection passport

Every investigator keeps a record of how their own thinking changed. Stamp each page of your passport — one honest sentence per stamp is enough.

HOW YOUR PASSPORT WILL BE STAMPED

Stamp level	What it looks like
Boarding pass	One stamp filled in; the rest still blank — the journey started, and that counts.
Domestic	Three stamps with honest one-line sentences; the words are your own.
International	All five stamps plus the final page — and every sentence names a machine, a limit or a fix.
Around the world	Everything above, plus one stamp that your classmate reads aloud and agrees truly describes them too.

T7-17 Reflection Passport

ALONE · 10 MIN

STAMP U · UNDERSTAND

The three learning jobs, in my own words:

STAMP D · DATA

The chart trick I will never fall for again:

STAMP L · LEARN

What training and testing taught me about my own studying:

STAMP W · WORLD

One AI helper near me, its benefit and its limit:

STAMP R · RESPONSIBILITY

One fix I would fight for — and who must answer for it:

FINAL PAGE · THE BIG QUESTION

"How can a machine learn without being told the rules?" — my answer, with my reasons:

MACHINE MAINTENANCE LOG — REAL MACHINES GET SERVICED

Every learning machine has days when the world stops matching its pattern. Note the two moments your class-machine wobbled, and what fixed it — this is what AI teams call **maintenance**:

When it wobbled	What the world did that our pattern missed	What we changed or checked

GALLERY WALK · PEER REVIEW

Swap books with another team and study **their** machine. Real AI teams review each other's work — so do honest investigators.

The clearest chart they drew, and why:

One question I would ask their machine:

One thing their bias check caught that ours missed:

Give them one star and one wish: a thing they did brilliantly, and one thing to train better next time.

EXIT TICKET — BEFORE YOU CLOSE THE CASE

1. Name the three learning jobs in six words or fewer:

2. One chart trick I will now never fall for:

3. One question I will carry to Class 8:

Self-check

shade one: first circle = not yet · middle = almost · last = got it

- ☐ I collected structured data, drew both charts, and read a pattern from them with numbers ○○○
- ☐ I made a prediction from my pattern and said how sure I was ○○○
- ☐ I ran the bias check on my own sample — and said who was missing ○○○
- ☐ I graded my own machine honestly against the test set ○○○

THINK & INK

Working as the machine changed how I see real machines: the part I now respect most is … and the part I now question hardest is …

CHAPTER 5 CLUES — POCKET THESE

- The full loop: **collect → chart → pattern → predict → check the sample → grade honestly**.
- More data helps only when it is **more varied** — louder data can just repeat the mistake.
- Predicting a **number** is regression thinking: read the trend, then say **how sure** you are.
- A good machine — like a good investigator — **asks for help** when the world breaks its pattern.

The clue I would tell my family tonight:

Chapter 5 Checkpoint

one line each — no peeking back!

ALONE · 3 MIN

1. Say the survey-machine loop in order, without looking back. Which step do most people skip?

2. Our sample was all Class 7 cricket players. Which two groups were missing — and what could that do to the prediction?

3. The other section's top answer was NOT our prediction. Was our machine useless? What is the honest next step?

MY CASE JOURNAL — THE LAST CLUE OF CASE 2, FOR THE ROAD TO CLASS 8

My AI words

The fourteen words on this case — each one met through an experience, never before.

1 classification
Putting things into groups that already have names, like fruit or vehicles.
USE IT IN MY OWN SENTENCE:

2 regression
Using a pattern to predict a number, like a price or a temperature.
USE IT IN MY OWN SENTENCE:

3 clustering
Finding groups in examples that have no labels — the machine invents the groups.
USE IT IN MY OWN SENTENCE:

4 training set
The practice examples a machine learns from before it meets new ones.
USE IT IN MY OWN SENTENCE:

5 test set
The hidden examples used to check whether the learning really worked.
USE IT IN MY OWN SENTENCE:

6 feature
A useful clue, stored as a number, that helps a machine decide.
USE IT IN MY OWN SENTENCE:

7 frequency
How many times a word appears — the number a counting machine reads.
USE IT IN MY OWN SENTENCE:

8 recommendation
A suggestion made by finding people whose pattern of likes matches yours.
USE IT IN MY OWN SENTENCE:

9 structured data
Facts arranged in rows and columns, so patterns can be found.
USE IT IN MY OWN SENTENCE:

10 trend
The general direction a line graph shows: rising, falling or steady.
USE IT IN MY OWN SENTENCE:

11 sample
The group you actually collected data from — always smaller than everyone.
USE IT IN MY OWN SENTENCE:

12 bias
When examples are one-sided, so the machine's results treat some people unfairly.
USE IT IN MY OWN SENTENCE:

13 misinformation
False or edited content that spreads — sometimes by mistake, sometimes on purpose.
USE IT IN MY OWN SENTENCE:

14 accountability
Being answerable for what a system does — only people can hold it.
USE IT IN MY OWN SENTENCE:

DETECTIVE'S OATH

Machines **sort, count, predict** — people **choose the examples, check the sample and stay answerable**. When a machine speaks, I will ask my four questions: **How do I know? What is missing? Who made this? What is the trick here?**

ANSWER HINTS — FOR CHECKING, NEVER FOR COPYING

T7-02: the best line is close to ₹25 plus ₹14–15 per km; a 6.5 km trip lands near ₹115–125; the 12 km guess is riskier because it stretches beyond every dot you saw. · **T7-04 exam set:** IN = 27, 36, 45; OUT = 14, 19. · **T7-07:** the counting rule flags messages 1, 3 and 5. · **T7-11:** Chart A is honest; Chart B's axis starts at 88 instead of 0; Chart C's shares add up to 110%. · **Ch 1 drill, in order:** C, N, G, C, N, G. · **Ch 2 drill:** reader, recommender, camera, recommender, camera, reader. · **Ch 3 drill:** hottest is May; rising from Jan to May, dipping in Jun.

MORE HINTS — FOR THE STICKY MISSIONS

T7-03: a group of fliers, a group of swimmers and a group of walkers are all honest groupings — remember, clustering has no single right answer. · **T7-06:** watch-words for the camera: corners, edges, brightness patches. · **T7-08:** "people like you" usually means the two or three rows with the most matches in the LIKE column, not your best friend. · **T7-10:** every helper is **data in → prediction out → human decides** — if your report card misses the human, it is not finished. · **T7-13:** look at what the Uniform Suggester was shown, never at what you wish it had been shown. · **T7-16 Step 6:** the honest answer is almost never "no bias" — name **which** class or street is missing.

MY HINT LOG — WHICH MISSIONS NEEDED A PEEK?

Hints are training data for you. Tick the mission code each time a hint unstuck you — a mission that needed three hints deserves one more retry next week, not three more hints:

- ☐ T7-__ needed a hint
- ☐ T7-__ needed a hint
- ☐ T7-__ needed a hint
- ☐ T7-__ needed a hint

A POSTCARD TO THE CLASS 8 INVESTIGATOR — WHO WILL BE YOU

The next case asks: **"What does it take to build an AI that people can trust?"** Leave three clues for your future self — the machine that impressed you most, the mistake that taught you most, and one question you dare Class 8 to answer:

My progress

Shade honestly — investigators never fake evidence.

MY OUTCOME TRACKER — TICK WHEN YOU TRULY CAN DO IT

- ☐ 7.U1 — tell classification, regression and clustering apart, with classroom examples
- ☐ 7.L1 — explain training vs testing as "practice questions vs the real exam"
- ☐ 7.L2 — explain why few or one-sided examples fool a learner, and how varied ones help
- ☐ 7.D1 — collect structured data; draw and read bar, line and pie charts; spot a misleading one
- ☐ 7.W1 — explain at idea level how a machine sees, reads and recommends
- ☐ 7.W2 — describe AI uses in healthcare, education, transport, agriculture, communication — a benefit and a limit each
- ☐ 7.R1 — explain how biased examples cause unfair results, and propose a fix
- ☐ 7.R2 — apply the three checks — source, evidence, emotion — to fake or edited content
- ☐ 7.T1 — compare solving a task with fixed if-then rules vs learning from examples

MY CASE FILES — SHADE A STAR PER CHAPTER FINISHED

1 Learning Machine · 2 See & Read · 3 AI at Work · 4 Gets It Wrong · 5 Survey Machine



PRATIMAI · AI HANDOUTS · CLASS 7

My Machine-Trainer Certificate

This certifies that investigator

_____

INVESTIGATOR NAME

has completed the case file **"How Machines Learn"** — Level 2 · SORT & PREDICT — solved all seventeen missions, and promised to keep asking: **How do I know? · What is missing? · Who made this? · What is the trick here?**

_____	_____	_____
DATE	INVESTIGATOR'S SIGNATURE	GUIDE'S SIGNATURE

Ready for the next case? Class 8 asks: **"What does it take to build an AI that people can trust?"** — bring your badge.